

Unit Conventions and the Speed of Light  $c$   $E=mc^2$  vs.  
 $E=m$ : Two Equivalent Perspectives Natural Units, SI Units, and the T0  
Viewpoint

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**Correction notes**

**Note on the corpus status: variable speed of light.** The idea of a varying speed of light referred to in this document has been replaced in the later corpus by *fractal path lengthening*: the geometric stretching of the propagation path in the  $\xi$  field (Doc. 026, 149, 182; achromatic according to K6 in Doc. 190). Statements that rely on a variable speed of light are to be read accordingly.

**Corrections from the calculation check (30 Sept. 2026).** The following places are corrected or annotated in the text; each carries a dated note with the previous value:

- Lorentz transformation of energy in SI:  $E' = \gamma(E - p \cdot v \cdot c^2) \rightarrow E' = \gamma(E - v p_x)$ .
- $c = L/T \propto (1/m)/(1/m) = \text{"scale-dependent"}: \text{annotated} - \text{the quotient is mass-independent.}$

**Abstract**

This document examines when one can set  $c=1$  (natural units) and when the full form  $E=mc^2$  with  $c=299,792,458$  m/s (SI units) is required. Parallel to the treatment of the fine-structure constant  $\alpha$  in Document 101, we show: Both perspectives are mathematically equivalent and differ only in the choice of unit system. The T0 theory reveals that  $c$  is not a fundamental law of nature but a dynamic ratio  $L/T$ . From the T0 perspective,  $c=1$  can be set (Planck units, particle physics), while for technical applications and precision measurements, SI units with explicit  $c$  are required. The equivalence  $E=mc^2 \leftrightarrow E=m$  holds exactly in natural units. References: Documents 013 (SI system), 014 (nat./SI), 015 (systematics), 077 ( $E=mc^2$  analysis), 101 ( $\alpha$  conventions).

**.1 Introduction: The Question of  $c=1$** **The Central Question**

The question "When can one set  $c=1$ ?" is analogous to the question "When can one set  $\alpha=1$ ?" addressed in Document 101. In both cases, it concerns **unit conventions**, not fundamental physics.

**Central Thesis**

**$E=mc^2$  and  $E=m$  are mathematically identical!**

- In SI units:  $E = mc^2$  with  $c = 299,792,458$  m/s
- In natural units:  $E = m$  with  $c = 1$
- Table: " $\alpha = 1$  (Heaviside-Lorentz)"  $\rightarrow$  HL with  $e^2 = 4\pi$ ; in HL alone  $\alpha = e^2/4\pi \approx 1/137$ . Both forms describe the same physics – only the unit choice differs.

**Historical Context**

Einstein wrote the famous formula in 1905:

$$E = mc^2 \quad (1)$$

In 1905 Einstein worked in the Gaussian/cgs framework (the SI exists only since 1960) and wrote the result as  $m = L/V^2$ . The form  $E = mc^2$  is needed in every unit system in which

length, time and mass are independent dimensions – today in the **SI** with meter, second and kilogram.

**Modern particle physics** uses instead:

$$E = m \quad (\text{in natural units with } c = \hbar = 1) \quad (2)$$

## .2 Natural Units: When $c=1$ is Valid

### Definition of Natural Units

In natural units, one sets:

$$c = 1, \quad \hbar = 1, \quad (\text{optional: } k_B = 1) \quad (3)$$

**Mathematical meaning:**

$$c = 1 \quad \Rightarrow \quad \text{Length} \equiv \text{Time} \quad (4)$$

$$\hbar = 1 \quad \Rightarrow \quad \text{Energy} \equiv \text{inverse Time} \quad (5)$$

### Application Domains

**Natural units are appropriate in:**

- **Planck scale:** Quantum gravity, fundamental theory
  - **Particle physics:** High-energy physics, QFT, Standard Model
  - **Cosmology:** Early universe, inflationary models
  - **Theoretical work:** Mathematical derivations, symmetries
- Advantage:** Formulas become simpler, physical relationships clearer.

### Mathematical Consistency

In natural units:

$$E^2 = p^2 + m^2 \quad (6)$$

In the rest frame ( $p = 0$ ):

$$E = m \quad (7)$$

This is exact – **not an approximation**.

### T0 Perspective: $c$ as a Ratio

The T0 theory shows (see Document 077):

$$c = \frac{L}{T} \quad (8)$$

**$c$  is not a fundamental law of nature but a *ratio*!**

With the T0 fundamental relation:

$$T \cdot m = 1 \quad (\text{Time-Mass Duality}) \quad (9)$$

it follows that  $c$  is a dynamic ratio that varies with mass scale.

**Implication:** In Planck units, where  $t_P = \ell_P/c$ ,  $c=1$  is the natural choice.

### .3 SI Units: When $c=299,792,458$ m/s is Required

#### The SI Definition (since 2019)

The modern SI system defines since 2019:

$$c = 299,792,458 \text{ m/s (exact)} \quad (10)$$

This choice is a **convention** that defines the meter via the second.

#### Application Domains

SI units with explicit  $c$  are required in:

- **Engineering:** GPS, telecommunications, laser technology
  - **Precision measurements:** Atomic clocks, interferometry, metrology
  - **Experimental physics:** Laboratory measurements with SI-calibrated devices
  - **Applied physics:** Energy calculations, dosimetry
  - **Public & Education:** Comprehensibility, historical continuity
- Advantage:** Practical calculability with calibrated measurement devices.

#### Mathematical Form

In SI units:

$$E = \gamma mc^2 \quad (11)$$

with the Lorentz factor:

$$\gamma = \frac{1}{\sqrt{1 - v^2/c^2}} \quad (12)$$

In the rest frame ( $v = 0, \gamma = 1$ ):

$$E = mc^2 \quad (13)$$

#### Conversion Between Unit Systems

From natural units to SI:

$$E_{\text{nat}} = m_{\text{nat}} \quad (14)$$

$$\Downarrow \text{ (Multiply by } c^2 \text{)} \quad (15)$$

$$E_{\text{SI}} = m_{\text{SI}} \cdot c^2 \quad (16)$$

**Example:** Electron mass

$$m_e = 0.511 \text{ MeV (natural units)} \quad (17)$$

$$m_e = 9.109 \times 10^{-31} \text{ kg (SI)} \quad (18)$$

$$E_e = m_e c^2 = 0.511 \text{ MeV} = 8.187 \times 10^{-14} \text{ J} \quad (19)$$

| Convention           | Fine-structure constant $\alpha$                    | Speed of light $c$       |
|----------------------|-----------------------------------------------------|--------------------------|
| <b>Natural</b>       | $\alpha = 1$ (Heaviside-Lorentz with $e^2 = 4\pi$ ) | $c = 1$ (Planck units)   |
| <b>SI / Standard</b> | $\alpha = 1/137.036$ (Gauss-SI)                     | $c = 299,792,458$ m/s    |
| <b>Document</b>      | 101 (Circularity-Constants)                         | 134 (Unit Conventions c) |

**Table 1:** Parallel structure:  $\alpha$  and  $c$  as conventions

(Corrected on 30 Sept. 2026: previously " $\alpha = 1$  (Heaviside-Lorentz)" – in Heaviside-Lorentz units alone  
 $\alpha = e^2/4\pi \approx 1/137$ ;  $\alpha = 1$  additionally requires  $e^2 = 4\pi$ .)

## .4 Comparison with $\alpha$ : Parallel Structure

### Two Analogous Conventions

#### Common Principles

Both cases show:

- **Physics is invariant** under unit choice
- **Natural units** simplify theoretical work
- **SI units** enable practical applications
- **T0 theory:** Both are derived conventions, not fundamental

#### T0 Reduction

From the T0 perspective (see Document 101):

$$\xi \rightarrow D_f \rightarrow E_0 \rightarrow \alpha \rightarrow \hbar, c, G \rightarrow \text{all other constants} \quad (20)$$

**Only  $\xi = \frac{4}{3} \times 10^{-4}$  is fundamental.**

Both  $\alpha$  and  $c$  are derived quantities or conventions.

## .5 When to Use Which System?

### Decision Matrix

| Context                | Natural units ( $c=1$ ) | SI units ( $c$ explicit) |
|------------------------|-------------------------|--------------------------|
| Theoretical physics    | ✓                       |                          |
| Quantum field theory   | ✓                       |                          |
| High-energy physics    | ✓                       |                          |
| Early cosmology        | ✓                       |                          |
| Experimental physics   |                         | ✓                        |
| Engineering            |                         | ✓                        |
| Precision measurements |                         | ✓                        |
| Applied physics        |                         | ✓                        |
| Education              |                         | ✓                        |

**Table 2:** Application domains of unit systems

## Recommendations

### Use natural units ( $c=1$ ) when:

- Performing theoretical derivations
- Symmetries and invariant structures are important
- Formulas should be simplified
- Working in particle physics or cosmology

### Use SI units ( $c$ explicit) when:

- Planning or evaluating experimental measurements
- Technical calculations are required
- Results should be understandable for non-physicists
- Historical continuity is important

## .6 Common Misconceptions

### “ $c=1$ is only an approximation”

**FALSE.**  $c=1$  is **exact** in natural units, not an approximation.

It is a choice of unit system that defines:

$$\text{Length unit} = \text{Time unit} \quad (21)$$

Analogously: In Planck units,  $\hbar = 1$  is exact, not approximate.

### “ $E=m$ only holds for photons”

**FALSE.** In natural units,  $E = m$  holds for **all** particles in their rest frame.

For photons ( $m = 0$ ):  $E = p$  (in natural units) or  $E = pc$  (in SI).

### “ $c$ is a fundamental constant of nature”

**T0 viewpoint:**  $c$  is a **ratio**  $L/T$ , not a fundamental constant.

With the T0 duality  $T \cdot m = 1$ ,  $c$  varies dynamically with mass scale:

$$c = \frac{L}{T} = L \cdot m \quad (22)$$

Only in SI units is  $c$  *fixed by definition*.

### “Natural units change the physics”

**FALSE.** Physics is independent of the unit system.

All **dimensionless** quantities (e.g.,  $\xi$ ,  $\alpha$ , mass ratios) are invariant.

Only dimensional quantities change their numerical values.

## .7 T0 Perspective: c as a Dynamic Ratio

### The T0 Fundamental Relation

From Document 077:

$$T \cdot m = 1 \quad (\text{Time-Mass Duality}) \quad (23)$$

This means:

$$T \propto \frac{1}{m} \quad (24)$$

$$L \propto \frac{1}{m} \quad (\text{via Compton wavelength}) \quad (25)$$

$$\Rightarrow c = \frac{L}{T} \propto \frac{1/m}{1/m} = \text{scale-dependent} \quad (26)$$

(Note of 30 Sept. 2026: The quotient  $(1/m)/(1/m) = 1$  is mass-independent; the calculation shows that  $c$  stays the same across all mass scales, not that it is scale-dependent. The conclusions below (different effective  $c$  values at different mass scales) and  $c = L \cdot m$  in the section on misconceptions rest on this line.)

### Implications

#### 1. c is not universally constant in the T0 framework:

Different effective  $c$ -values can occur at different mass scales.

#### 2. SI definition $c=299,792,458$ m/s is a calibration:

This fixation defines the meter via the second – a metrological convention.

#### 3. Natural units $c=1$ are T0-consistent:

In Planck units, where  $t_P \propto \ell_P$ ,  $c=1$  is the natural choice.

### Comparison with Document 077

Document 077 argues: “ $E=mc^2 = E=m$  – The Constant Illusion Exposed”

#### Clarification here:

- $E=mc^2$  (SI) and  $E=m$  (natural) are *equivalent*, not identical
- The difference lies in the *unit system*, not in physics
- Einstein's  $c$ -fixation is a *convention*, not an error
- T0 shows:  $c$  is a ratio that can vary depending on scale

## .8 Mathematical Consistency

### Energy-Momentum Relation

In natural units ( $c = 1$ ):

$$E^2 = p^2 + m^2 \quad (27)$$

In SI units:

$$E^2 = (pc)^2 + (mc^2)^2 \quad (28)$$

Both forms are mathematically equivalent.

## Lorentz Transformation

In natural units:

$$E' = \gamma(E - p \cdot v) \quad (29)$$

In SI units:

$$E' = \gamma(E - v p_x) \quad (30)$$

(Corrected on 30 Sept. 2026: previously  $E' = \gamma(E - p \cdot v \cdot c^2)$  – with  $p$  in kg·m/s and  $v$  in m/s,  $p v$  already has the unit J; the factor  $c^2$  would give J·m<sup>2</sup>/s<sup>2</sup>. In SI the form is the same as in natural units.)

The physics remains invariant.

## Klein-Gordon Equation

In natural units:

$$(\partial_\mu \partial^\mu + m^2)\phi = 0 \quad (31)$$

In SI units:

$$\left( \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \nabla^2 + \frac{m^2 c^2}{\hbar^2} \right) \phi = 0 \quad (32)$$

Identical physics, different notation.

## .9 References to T0 Documents

### Related Documents

- **Document 013:** SI System and T0 Theory
- **Document 014:** Natural vs. SI Units
- **Document 015:** Systematics of Natural Units
- **Document 077:**  $E=mc^2 = E=m$  Analysis
- **Document 101:** Circularity of Constants ( $\alpha$  Conventions)
- **Document 133:** Fractal Correction  $K_{\text{frak}}$  Derivation

### Derivation Hierarchy

The T0 hierarchy (from Document 101):

$$\xi \rightarrow D_f \rightarrow E_0 \rightarrow \alpha \rightarrow \hbar, c, G \rightarrow \text{mass ratios} \quad (33)$$

shows that both  $\alpha$  and  $c$  are derived quantities.